

Single Source Justification (Simplified Acquisition)

Section A: General Contract Information

Is this a Bridge Action as defined in the [AF Bridge Action Reduction Plan](#)? ☐ Yes ☒ No

Contracting Activity: AFTC/PZIA (Arnold)

Project/Program Name: National Full-Scale Aerodynamics Complex (NFAC) Switch Matrix

Purchase Request / Local ID Number: [REDACTED]

Estimated Contract Cost (including Options): [REDACTED]

Section B: Description of the Supplies/Services Required

The requirement is for two Precision Filters, Inc., 464kC 256x256 Solid State Switch Matrix Systems to support the National Full-Scale Aerodynamics Complex (NFAC) operations. Each complete system consists of:

- Qty 1 - 464kB-R1 Mainframe with Front Panel, Controls and Displays, Midplane and Labeling
- Qty 1 - 464kD-ACPS Power Supply
- Qty 1 - 464kC-C5-S1T Control with Ethernet
- Qty 1 - 464k-GUI-WIN Graphical User Interface for 464K System
- Qty 1 - 464k-TEST-WIN Factory Acceptance Test Software
- Qty 8 - 464kC-I-MP2-(2)16x31 Input Card, 32 Shielded Pairs
- Qty 8 - 464kC-O-MP2-(2)31x16 Output Card, 32 Shielded Pairs
- Qty 8 - 464kB-INTER-(4)16x16 Interstage Switch Module
- Qty 1 - SM24B Slide Mount Kit, 24" for 464k Mainframe
- Qty 2 - 464k-I/O-TEST Test Cable (2 required per system for FAT test)
- Qty 16 - 464k-I-SHORT Input Shorting Test Adapter (2 req'd per input card for FAT test)
- Qty 32 - CONN-OUT-50D-A 50-Pin Mating Output Connector with Crimp Pins for Output Connectors on MF with 0.59" diameter cable entry (A5092G5)

This action is for a new Single-Award Indefinite-Delivery Indefinite-Quantity (IDIQ) contract to Precision Filters, Inc., for brand name commercial items. The contract will be awarded as a firm-fixed-price contract under the authority of FAR 13.106-1(b)(1)(i). The IDIQ contract will have a minimum guarantee of one complete switch matrix system and a maximum quantity of two complete systems.

The requirement is for Precision Filters, Inc., 464kC 256x256 Solid State Switch Matrix Systems to support NFAC operations. One system will be procured in the current fiscal year through an initial delivery order, with a second system planned for procurement in Year 3 through a subsequent delivery order. The total estimated value is [REDACTED] (per system). The total estimated value for both systems is [REDACTED] with each system valued at [REDACTED]. The first system will be procured in the current fiscal year with delivery required 120-150 days after receipt of initial delivery order. The second system is planned for procurement in Year 3, also requiring 120-150 days for delivery after receipt of order. This requirement will be funded using 3600 (Research, Development, Test & Evaluation, Air Force) funds for FY 25.

This justification covers a multi-year requirement due to the planned procurement schedule and the need to maintain guaranteed pricing for both systems. The IDIQ contract structure will establish firm-fixed unit pricing for both systems at time of award. The maximum dollar value of [REDACTED] represents the total estimated value for both systems over the contract period. All ancillary requirements, including technical documentation, testing software, and support equipment, are included in the complete system configuration. No additional requirements beyond these components will be procured under this contract.

Section C: Justification for Soliciting from a Single Source or Brand Name

The NFAC currently operates two Precision Filters, Inc., switch matrices, one in each wind tunnel. This acquisition requires exact compatibility with these existing systems to support the development of a new data system at AEDC. The brand name requirement for Precision Filters, Inc., switch matrix systems is justified based on both technical requirements and substantial

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cost savings to the Government.

Technical compatibility is critical as these systems must integrate seamlessly with existing NFAC infrastructure. The 464kC 256x256 Solid State Switch Matrix Systems provide specific technical capabilities that are essential for NFAC operations:

- Precise signal routing through 256x256 crosspoint switching architecture
- High-density I/O capability with 32 shielded pairs per card
- Specialized interstage switching modules for complex signal paths
- Proprietary control interface protocols that integrate with existing NFAC systems
- Custom GUI software that maintains operational consistency across facilities
- Factory acceptance testing protocols specific to NFAC requirements
- Unique midplane architecture ensuring signal integrity across all channels
- Specialized shielding requirements for wind tunnel testing environments

The technical architecture of these systems is specifically designed for wind tunnel testing applications, with features such as:

- Low-noise signal paths critical for accurate data collection
- High-speed switching capabilities necessary for real-time testing
- Integrated system monitoring and diagnostic capabilities
- Specialized grounding schemes for test environment compatibility
- Custom connector configurations matching existing infrastructure

The Government would need to procure four new switch matrices instead of two if the requirement is not satisfied by brand name basis, as the current plan makes use of existing hardware. The procurement cost would double from [REDACTED] to approximately [REDACTED]. The brand name justification, therefore, represents a significant cost savings by allowing continued use of existing hardware while maintaining system consistency.

The requirement for Precision Filters, Inc., systems is not due to lack of advance planning but rather a deliberate technical decision based on integration requirements with existing NFAC infrastructure, prevention of substantial duplication of costs, maintenance of operational continuity across facilities, support for ongoing development of new NFAC data system, ensuring availability of operational spares, preservation of established calibration and testing protocols, maintenance of existing signal routing architectures, and continuation of proven system reliability in wind tunnel environments.

This justification supports a single contract action and does not include any unpriced options. All pricing will be established at time of contract award for both potential delivery orders. This is not a class justification as it applies to a single IDIQ contract with Precision Filters, Inc.

The requirement meets the conditions for simplified acquisition procedures for certain commercial items, and the brand name specification is necessary to satisfy the agency's requirements for compatibility with existing NFAC infrastructure.

The statutory authority permitting other than full and open competition for this acquisition is DAFFARS 5313.501 - 10 U.S.C. 3572, Implementation of Simplified Acquisition Procedures (41 U.S.C. 1901). This authority is appropriate as this is a commercial item acquisition below the Simplified Acquisition Threshold using brand name justification under FAR 13.106-1(b)(1)(i). The requirement meets the conditions for simplified acquisition procedures for certain commercial items, and the brand name specification is necessary to satisfy the agency's requirements for compatibility with existing NFAC infrastructure.

Section D: Efforts to Obtain Competition

The Notice of Intent clearly describes the Government's requirements for Precision Filters, Inc., 464kC 256x256 Solid State Switch Matrix Systems and provides potential sources the opportunity to submit their capabilities if they believe they can meet the requirement. The notice includes detailed specifications of the required systems and emphasizes the critical need for complete compatibility with existing NFAC infrastructure.

Market research was conducted through various methods including internet research, review of similar electronic testing equipment manufacturers, and direct communication with the original equipment manufacturer. The research confirmed that

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while other manufacturers produce switch matrix systems, none can provide the exact compatibility required without necessitating complete replacement of existing NFAC infrastructure, which would result in substantial duplication and increased cost.

No qualifying country sources have expressed interest in this requirement. The commercial nature of these items and the brand name justification based on compatibility requirements make consideration of alternative sources impractical, as it would require complete replacement of existing infrastructure and result in doubled procurement costs.

The Government has taken appropriate steps to ensure maximum practicable competition while balancing the critical need for system compatibility and cost efficiency. The use of simplified acquisition procedures and the posting of a Notice of Intent demonstrate the Government's commitment to transparency in this procurement action while maintaining the technical requirements necessary for successful mission accomplishment.

Section E: Steps to Preclude Future Single Source or Brand Name Awards

This justification is supported by market research which provided potential sources the opportunity to demonstrate their capability to meet these requirements. The use of simplified acquisition procedures is appropriate given the commercial nature of these items and the total value being below the Simplified Acquisition Threshold.

Section F: Contracting Officer's Determination

I have determined, in accordance with [FAR 13.106-1\(b\)\(1\)](#), that the circumstances of this contract action deem only a single source is reasonably available.

Date 23-Apr-2025	Typed Name and Rank/Grade of Contracting Officer LESLIE R. MCGOWAN, GS-12 Contracting Officer	Signature of Contracting Officer MCGOWAN.LESLIE.RENEE.1 141842702  Digitally signed by MCGOWAN.LESLIE.RENEE.141842702 Date: 2025.04.23 13:42:11 -05'00'
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